

Large-Scale Plastic Detection Pipeline via Machine Learning-Augmented Remote Sensing

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Abstract

Plastic pollution continues to escalate globally, yet the large-scale detection of terrestrial plastics remains limited by inefficient manual sampling. Prior studies have used remote sensing for particulate and debris detection, but none have isolated optimal spectral bands for plastic identification in terrestrial settings. This study proposes a machine learning-augmented remote sensing pipeline to optimize detection accuracy and scalability. In Phase I, five spectral bands – near-infrared (NIR), red-edge, red, green, and visible – were evaluated using drone imagery collected over terrestrial backgrounds. The plastic litter used were chosen for their compositions of high-frequency litter: High-Density Polyethylene (HDPE) and Polyethylene Terephthalate (PET). An SSD-MobileNet v3 model served as a machine learning proxy to compare effectiveness scores across bands. Contrary to previous literature, results showed that the green (560 nm) band outperformed NIR, achieving the highest object-detection effectiveness (* $p < .05$). The green band's dominance was attributed to greater plastic-to-background pixel contrast, caused by the photochemical properties of plant matter, HDPE, and PET. In Phase II, the identified optimal band was leveraged to the full detection framework development, using the state-of-the-art RF-DETR_medium model. Trained and validated on 1,754 images, the pipeline achieved a precision of 88.8%, recall of 83.0%, and F1 score of 0.86, surpassing the object-detection proficiency benchmark of 0.85. These findings establish a viable, high-accuracy workflow for large-area macroplastic detection in terrestrial backgrounds. This research advances the automation of terrestrial plastic mapping; it lays the groundwork for real-time pollution monitoring systems capable of reducing cleanup inefficiency and supporting data-driven environmental management.

Introduction

Microplastic pollution is rising drastically. The particulates collect in water bodies, where they serve as disease vectors and damage local ecosystems. While it is important to remove microplastics from the environment, it is difficult to collect data on microplastic presence over large areas, preventing collection efforts from reaching high efficiency or low cost. Macroplastic presence data is more useful, as the larger objects are easier to detect, and they break down into microplastics through erosion. It is also more effective to tackle collection in terrestrial environments, as it sources over 60% of marine macroplastic pollution. Remote sensing is one method of aerial imagery that specializes in large-area data collection tasks, but an optimized sensing system for macroplastic litter in terrestrial environments has not yet been constructed. In this study, I created such a system. Phase I involves identifying the spectral wavelength that results in images best suited for plastic detection. This finding was coupled with specialized object detection machine learning in Phase II to create a robust detection system.

Review of Literature

Microplastics

Rising macroplastic production has made microplastic pollution ubiquitous. Erosional forces fragment large objects into smaller plastic particulates. Microplastics – particulates under 5 μ m (Wright et al., 2013) – disperse through air, land, and water, with marine microplastics varying in placement based on wind, salinity, and seafloor topography (Tsuchiya et al., 2023). While some particulates settle in sediments or within the water column, most reside on the surface. On average, individual humans ingest 20mg to 27mg of microplastic particulates annually (Cox et al., 2019), exposing them to carcinogens like phthalates, as well as pathogens and parasites that may be attached to the small pieces (Barboza et al., 2018; Vethaak & Leslie, 2016; Wu, 2017). These factors increase heart attack and stroke risk by up to 450% (Raffaele Marfella et al., 2024).

Cleanup efforts are hindered by limited data on microplastic distribution. Current methods of measuring microplastic concentration revolve around *in-situ* sampling. The method is effective for determining plastic composition: the most common polymers in plastic litter are high-density polyethylene (HDPE) and polyethylene terephthalate (PET) (Gandhi et al., 2021). However, the effectiveness of in-situ sampling is limited at scale: it risks inefficiency due to contamination and reliance on manual analysis, especially for large areas (Winans et al., 2023). To solve this issue, researchers turn to remote sensing, macroplastic pollution, and terrestrial environments.

Remote Sensing and Spectral Reradiation

Remote sensing analyzes reflected or emitted light across spectral bands using satellite or drone imagery. For example, fire scar analysis leverages the shortwave infrared band, in which the contrast between burned and unburned land is most pronounced (Sivakumar et al., 2003). This analysis of wavelengths that emphasize contrast or certain features is termed *Spectral Reradiation*, and it contributes to more insightful data trends within large-scale remote sensing efforts.

Spectral reradiation varies with surface composition and texture, as shown in Figure 1. One study used LANDSAT satellite data to optimize data collection of particulate pollution in lake surfaces. It revealed that R-band spectral reradiation best correlates with surface water particulate matter (Wen et al., 2023). However, it failed to distinguish plastics from other debris.

A study using remote sensing to map the Hawaiian coastline made use of machine learning to classify 80% of macro debris accurately, far surpassing manual analysis (Winans et al., 2023). Machine learning is an effective approach when targeting more efficient data analysis of macro-objects in terrestrial environments.

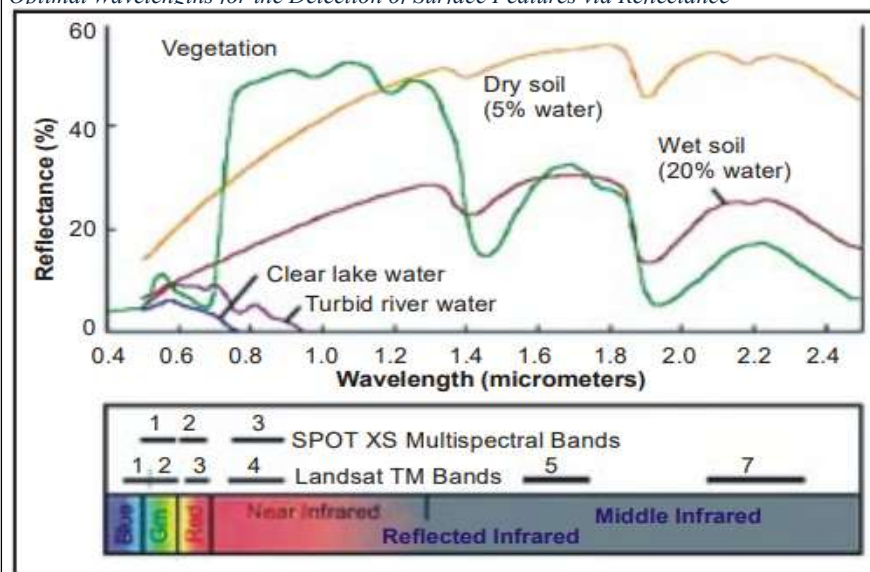
Utility of Machine Learning

Machine learning (ML) accelerates data analysis. Compared to manual efforts, ML has greater success when identifying data trends (Goyetche et al., 2023).

Specific model architecture types are more suited for specific types of data. Each model risks overfitting, a phenomenon where the system memorizes the data instead of learning; this results in

Figure 1

Optimal Wavelengths for the Detection of Surface Features via Reflectance



Note. Vegetation has the highest reflectance in 1.1 μm while turbid river water peaks at 0.6 μm .

Sivakumar, M., Roy, P., Harmsen, K., & Saha, S. (2003). *Satellite Remote Sensing and GIS Applications in Agricultural Meteorology* World Meteorological Organization (WMO) India Meteorological Department (IMD) Centre for Space Science and Technology Education in Asia and the Pacific (CSSTEAP) Indian Institute of Remote Sensing (IIRS) National Remote Sensing Agency (NRSA) and Space Application Centre (SAC). https://www.unisdr.org/files/1682_9970.pdf

poor applicability. Convolutional neural networks (CNNs) process image data by analyzing resolution, pixel values, and patterns. Due to their complexity, these require large datasets but are prone to overfitting (Winans et al., 2023). MobileNet, a CNN variant, reduces overfitting risk with a simpler architecture, making it effective for small datasets. The Detection Transformer (DETR) model, relatively new in the machine learning community, builds off of the CNN concept by employing a simplifying process called bipartite matching to streamline the system.

By choosing the appropriate ML architecture, a study would significantly increase the efficacy of data analysis. When applying a general model to a specific dataset, certain preparation and tuning steps are necessary to ensure high performance.

Implementing a Machine Learning Model

Implementation begins with acquiring and preprocessing data. Techniques such as jittering – adding image copies that are slightly shifted or warped to increase dataset size - and SMOGN – a method for correcting imbalances or skews within the sample data – are effective within preprocessing (Wen et al., 2023; Winans et al., 2023). The dataset is then split usually 80% for training and 20% for testing (Lin et al., 2024). Testing evaluates performance using F1 score, average precision, recall, and accuracy (Tedesco & Radzikowski, 2023).

Certain decisions must be made to achieve high accuracy while avoiding overfitting. The split percentages for training and testing are based on the task and the dataset. An object detection task with a very large dataset available may warrant a greater percentage for testing (such as 30%), because this would mean a more in-depth testing sequence, and thus more reliable results. However, such a split may not be sufficient for an object detection task with a much smaller dataset, resulting in a lesser testing proportion (such as 15%). Another decision is the number of epochs, or repetitions, a model should make. The choice is complex, and modern systems automate this through validation metrics calculated every N number of epochs. Once the validation loss consistently stops decreasing, training is stopped early. The goal for object detection is to achieve the accuracy threshold of 0.85 (Zhang et al., 2021; Tedesco & Radzikowski, 2023). Achieving this metric implies a sufficiently capable detection system for real-world application.

Problem Statements

PHASE I: We have yet to discover a correlation between electromagnetic spectrum reradiation from a terrestrial surface and macroplastic pollution presence.

PHASE II: Despite the presence of an optimal band, an efficient plastic detection pipeline utilizing large-scale remote sensing has not been developed.

Goals

PHASE I: To determine the correlation between specific spectral bands used in remote sensing imagery, and the ability of a machine learning system to detect and classify macroplastic objects on terrestrial surfaces.

PHASE II: To develop an efficient remote sensing detection pipeline focusing on terrestrial macroplastic objects.

Hypotheses

PHASE I: After comparing the effectiveness scores from each of five machine learning models – Near Infrared, Red-Edge, Red, Green, and Visible – the greatest effectiveness score will be demonstrated by NIR.

PHASE II: The performance of a full machine learning pipeline after tuning will surpass the object detection accuracy threshold of 0.85.

Methodology

Overview

Study Design

In order to tackle the plastic detection data gap, this research consisted of two phases. The work conducted in Phase I focused on optimizing the remote sensing approach to terrestrial plastic litter detection. I devised a *machine learning proxy* for this section. The comparison between bands would be based on the model results, isolating the most optimal spectral band without introducing bias. The work conducted in Phase II made use of the Phase I findings to create a robust plastic detection pipeline. In the context of this study, *pipeline* refers to the collection of steps between image acquisition and plastic identification.

I executed Phase I was during the fall of 2024; it focused on macroplastic objects within terrestrial environments. I executed Phase II during the summer of 2025; it focused on creating a robust plastic detection pipeline based on the findings of Phase I with regards to optimal data collection and analysis.

Role of Mentor

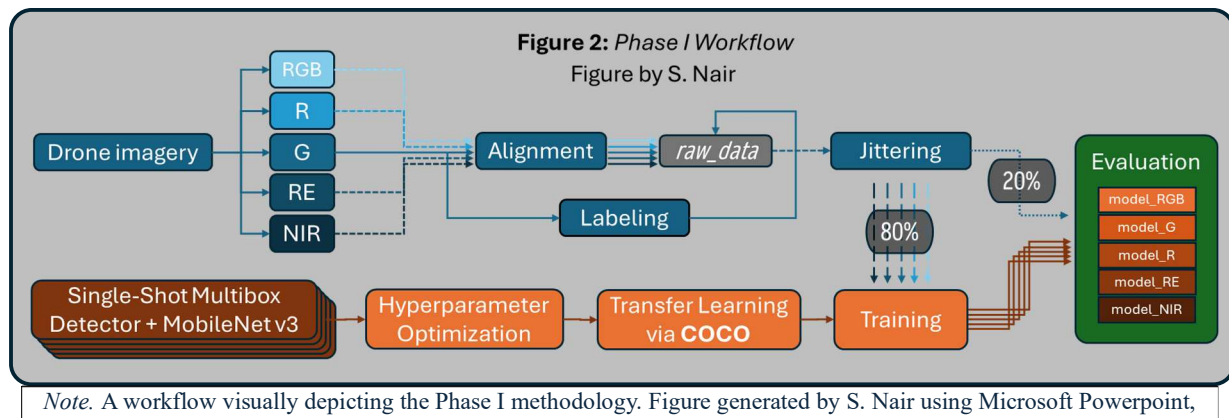
The role of the mentor was to provide feedback on feasibility and scope of my study design, given resource and time constraints. I performed all execution steps. The mentor assisted with some materials, including the DJI Mavic 3M drone; permissions to fly within the data collection area; transferring the image data from drone storage to a computer; and maintaining good drone condition across multiple days of data collection. The mentor also suggested certain machine learning modifications, which were developed in Phase II.

Data Management and Ethics

The data for both phases was processed through the Visual Studio Code and Windows 11 File Explorer applications. Details regarding application version, language version, and libraries utilized (all public access) can be found in Supplemental Details. In Phase II, the images were input into the RoboFlow online processing system to reduce time consumption and allow access to model training infrastructure.

Both phases of the project involved drone imagery of Earth surface. Some of the images contained instances of the student researcher and/or the mentor. To avoid privacy issues, all image chips containing such instances were discarded. No image was taken outside of the student researcher's personal computer and hard drive. Any data transfer utilized secure external databanks, USB drives, and/or Dropbox (a publicly available secure file share website).

PHASE I: Macroplastic Classification



Data Collection & Preprocessing

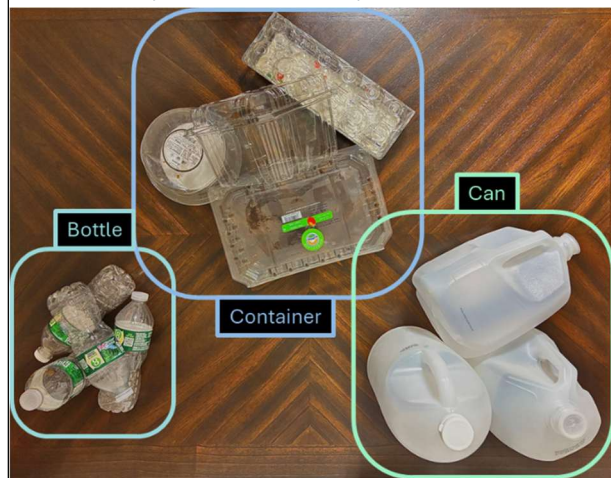
Collection Area. The collection area was located within the Lamont Doherty Earth Observatory in Palisades, New York. Two distinct backgrounds were utilized: an asphalt terrain, and a grass terrain.

Plastics Seeding. Macroplastics were sourced from local recycling receptacles. To test the detection system on common litter, HDPE and PET objects (Gandhi et al., 2021) were targeted. These were then sorted into three categories, or *classes*, visualized in Figure 3: Can, Bottle, Container. The Bottle class contained 53 PET plastic bottles. The Can class contained 136 HDPE water jugs and cooking oil containers. The Container class contained 34 PET plastic containers of varying shape.

The 223 macroplastic objects were seeded on the two backgrounds to mimic natural litter presence. Each background contained 10-15 “clumps” of overlapping objects, coupled with lone objects sparsely distributed. Approximately 50% of the clumps contained objects of uniform class, with the others having a mixture of two or three different classes.

Equipment. The drone used was a DJI Mavic 3M, equipped with visible light and multispectral cameras. The visible light camera included a 4/3 CMOS RGB image sensor with twenty million effective pixels, and a lens with 24mm focal length and adjustable focus. The multispectral camera included a 1/(2.8)-inch CMOS image sensor with five million effective pixels, and lens with 25mm focal length and fixed focus. The gimbal to which the cameras were affixed had an angular vibration range of ~ 0.007 , a vertical hovering accuracy of $\pm 0.1\text{m}$, and a horizontal hovering accuracy of $\pm 0.3\text{m}$. Figure 4 illustrates the camera positioning within the gimbal of the drone. The camera setup enabled image collection in four different

Figure 3
Visualization of Plastic Litter Classification Scheme



Note. A depiction of the three plastic litter classes used in this study: Can (left), Bottle (right), Container (center). Photograph taken and figure generated by S. Nair using Microsoft PowerPoint, 2025.

Figure 4
DJI Mavic 3M Camera Layout on Gimbal



Note. The NIR camera is slightly offset from the visible light (V) camera, labeled above. This necessitated image alignment post-data collection.

Adapted from DJI Mavic 3M Commercial Drone Platform, by DJI, 2024, Drone Nerds Enterprise (<https://enterprise.dronenerds.com/commercial-drone-platforms/mavic-series/dji-mavic-3m/>). Figure adapted and labeled by S. Nair using Microsoft PowerPoint, 2024.

spectral bands – 560nm (Green), 650nm (Red), 730nm (Red-Edge), and 860nm (Near-InfraRed). The first three bands had an error range of $\pm 16\text{nm}$, and the near infrared band had an error range of $\pm 26\text{nm}$.

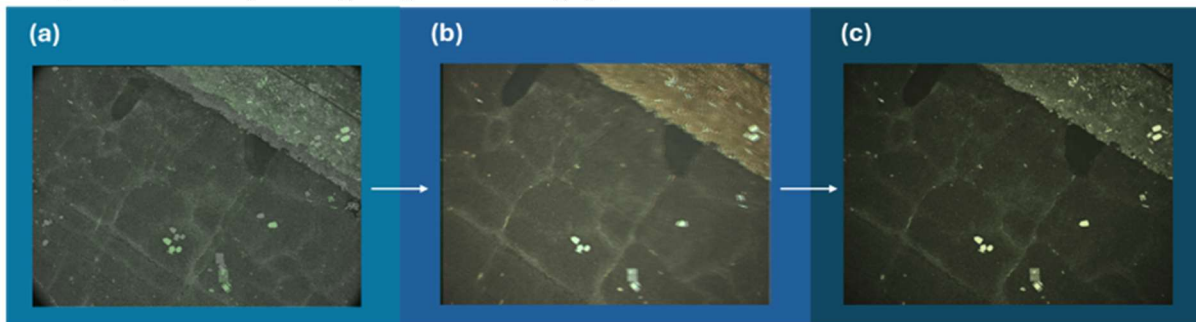
Drone Imagery. The drone was piloted by the student researcher to 50 meters above the Earth surface, capturing five image masks within each image – one per the four multispectral bands, and a fifth traditional visible light format. For clarity, each of these five datatypes are collectively referred to as *bands* within this study.

The raw image masks displayed misalignment due to camera positioning on the gimbal. Realignment correction was accomplished using a custom motion homography-based Python script, depicted in Figure 5. This preprocessing resulted in 134 raw images per dataset (per image band).

Labelling. Labelling was accomplished using labelImg, an open-source graphical image annotation tool. This was necessary to provide a comparison for the machine learning models. In each image within the Green band (chosen arbitrarily), all plastic objects were manually located and delineated with a *bounding box*, which stored the object's location and classification. A python script was executed to consolidate the resulting XML files into one CSV file containing each object's image source name, bounding box coordinates, and class.

Preprocessing. I developed UniChip, a custom preprocessing algorithm. First, the raw images were cropped, using an augmented sliding windows approach to avoid splicing plastic objects between multiple images. Then, new *image chips* were added via the jittering technique: by taking the existing image and applying rotations, linear shifts, and brightness and luminosity alterations, as depicted in

Figure 5
Image Alignment in Preprocessing Using Motion Homography



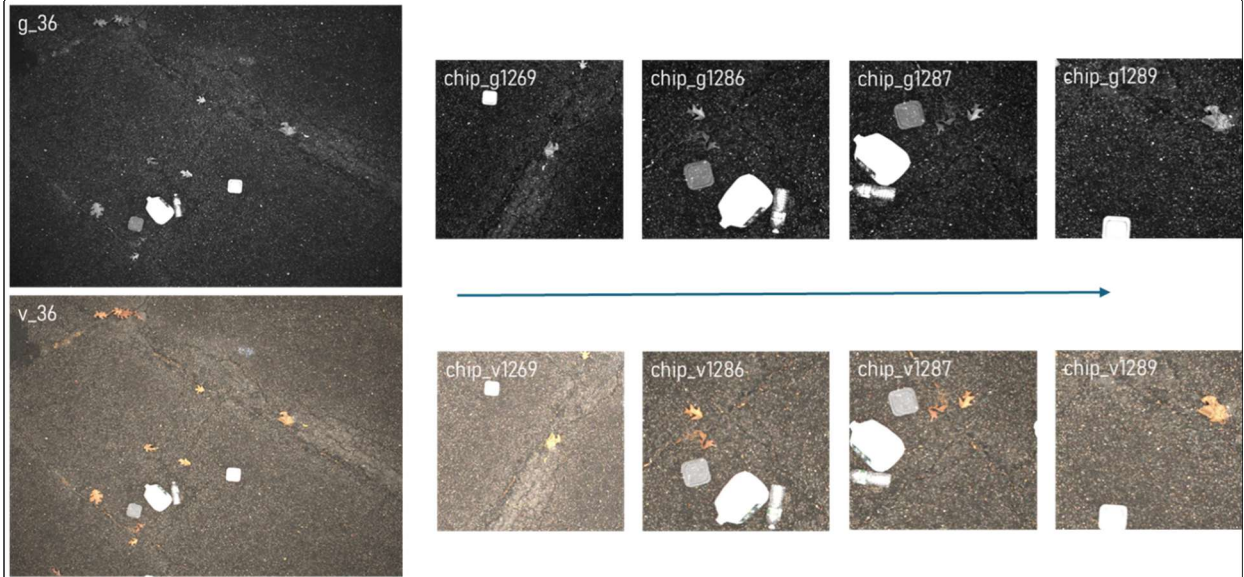
```
(a) image = Image.open(ROOT + "\\\" + f).copy()
      image.save(os.path.join(ROOT, 'new_data/raw_data', ["g", "nir", "r", "re"][sn%4] + "_" + str(sn//4) + ".tif"))

(b) image = image.crop((x1+crop_x_adjust, y1+crop_y_adjust, x2+crop_x_adjust, y2+crop_y_adjust))
      image = image.resize(target_size, Image.LANCZOS)

(c) _, warp_matrix = cv2.findTransformECC(rszImg1, rszImg2, warp_matrix, warp_mode, criteria, inputMask=None)
```

Note. (a) demonstrates the image alignment of g_1 and v_1 in their raw state. (b) demonstrates a significant reduction in misalignment, (c) has the greatest alignment between images. Figure generated by S. Nair using Microsoft PowerPoint, 2024.

Figure 6
Visual Depiction of UniChip Preprocessing Algorithm



Note. The corresponding chip generations on the right are locationally congruent, maintaining the congruency of the raw images shown on the left. Figure generated by S. Nair using Microsoft PowerPoint, 2025.

Figure 6. UniChip expanded the size of the dataset, working to improve model effectiveness, while maintaining congruency within corresponding images. This means that, when considering a corresponding set of image chips – one per each of the five bands – each chip contained the same plastics in identical locations, the only difference being the wavelength in which the chips were captured. This novel addition ensured that differences in model accuracy were solely due to wavelength. UniChip provided 694 image chips, which amounted to 3872 object instances, and 19360 training datapoints in total.

The train-test split was as follows: 70% of the chips were allocated for training, 30% for testing. This split was chosen based on Winans et al. (2023), who used the same model architecture on a similar task.

Machine Learning

Model. An SSD-MobileNet_v3 model sourced from Python’s Torchvision module was utilized; this is a combination of a Single-Shot Multibox Detector (SSD) and a MobileNet_v3 convolutional network, a system shown to be effective in debris classification (Winans et al., 2023). Five separate versions of the same model architecture were initialized and trained separately, one per band.

Training Infrastructure & Details. The models first underwent transfer learning via the Common Objects in Context (COCO) public dataset, ensuring general object detection capabilities.

Training parameters were identified through the work of Winans et al. (2023) or minor empirical testing. The initial learning rate was 0.005. The scheduler was “ReduceLROnPlateau”. The number of iterations was capped at 200 epochs, and a catch validation was included via loss analysis to exit if the model was displaying signs of overfitting. The validation loss was a 0.35-0.65 combination of generalized intersection-over-union (GIoU) loss – for location accuracy – and focal loss – for classification accuracy. It was calculated every third epoch; training would terminate if four consecutive epochs had unwanted (high) losses.

Model Evaluation & Statistical Analysis

Model Evaluation Metrics. As per the *machine learning proxy* scheme, the models were compared to isolate the most effective band. The *effectiveness* metric was created for this purpose; it is the weighted average of customary machine learning metrics, tuned to serve as an objective rating of model efficiency. The calculation can be found in Supplemental Details.

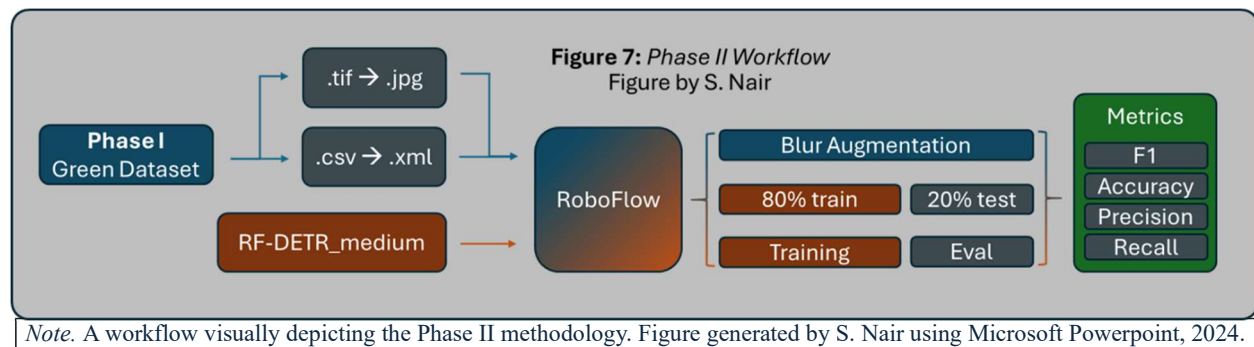
The model produced a confidence score from 0 to 1 for each of its predictions. A confidence score threshold needed to be determined, to remove low accuracy “bogus” predictions. To identify this threshold, the models were evaluated with all possibilities from 0.05 to 0.95 in increments of 0.05.

A set of paired t-tests were utilized to compare the confidence scores and determine the most effective model. To examine the contrast between background and plastics as a potential driver of model effectiveness, each dataset was analyzed by calculating the ratio between the average plastic pixel intensity and the average background pixel intensity.

Transition & Rationale

The work in Phase I focused on optimizing the plastic detection system. It identified the most optimal spectral band for the task, but it did not construct the system itself. Thus, Phase II concerned the creation of the pipeline, making use of the trends identified in Phase I.

PHASE II: Pipeline Construction



Data Collection & Preprocessing

The work conducted in Phase II utilized the five post-preprocessing datasets from Phase I. Based on the green band (560nm) advantage, only the Green dataset was used.

To utilize the RoboFlow scheme to boost training speed and model effectiveness, certain data corrections were necessary. As RoboFlow did not support the .tif image format, each of the Phase I image chips were converted to .jpg format. Next, to ensure that the Phase I green labels could be inputted into the RoboFlow system, XML files – used to “store” the information from the object labels – were created for each image. These two steps were executed via python scripts and the Phase I codebase.

RoboFlow provided certain preprocessing and jittering features similar to the methods used in the Phase I UniChip algorithm. One such feature was a blurring step. Image blurriness is a likely phenomenon when considering a simultaneous image capture during linear drone flight. Thus, the blur step was implemented into the preprocessing pipeline to ensure model robustness. In total, there were 1754 images with 9786 training datapoints.

Regarding the train-test split, RoboFlow provided a third denomination for validation. A specified validation dataset would result in a stronger model, due to more effective overfitting prevention and a more dynamic learning rate scheduler, boosting training. Thus, the split for train-validation-test was roughly 80% - 10% - 10%. However, after RoboFlow modified the groups to prevent biases such as class imbalance, the split was adjusted to 92% - 4% - 4%. The train dataset contained 1615 images, the validation dataset contained 70 images, and the testing dataset contained 69 images.

Machine Learning

Model Setup. To increase the efficiency of the pipeline, the system used RoboFlow’s RF-DETR model pretrained on the COCO dataset. This setup was chosen due to its versatility in object detection as a Detection Transformer architecture. Additionally, the specific combination is classified as a SOTA (state-of-the-art) model: it has been shown empirically to be the most successful framework in the domain of object detection and image segmentation.

Training Infrastructure. Training was executed with the RoboFlow cloud-computing GPU, boosting processing power. Regarding the hyperparameters, RoboFlow offered three systems with varying strengths. A common tradeoff in machine learning is accuracy with inference time: a model with greater accuracy generally is more complicated or nuanced and thus takes more time to output predictions. To determine the appropriate strength of the model used in this pipeline, I derived formulas solving for the maximum drone speed given the inference time of the model and factors impacting the

drone flight (see Supplemental Section). This was done to ensure that in a real-world scenario, the pipeline would be able to pair with the drone flight without introducing processing bottlenecks or data leakages.

The maximum speed of the DJI Mavic 3M drone is 15 meters per second; the height of the drone above the ground was 50 meters; the field of view of the green multispectral camera is 73.91°. Thus, the largest inference time that would not obstruct the drone’s flight would be 5.496 seconds. In comparison, the slowest RF-DETR setup (the medium variant) has an inference time of 183 milliseconds (0.183 seconds). This results in a max forward speed of 583 meters per second, or 1304 miles per hour – much faster than many drones can fly. Therefore, the RF-DETR_medium was used because it was the most accurate variant.

Model Evaluation & Statistical Analysis

Model Evaluation Metrics. The model was evaluated for accuracy using RoboFlow’s evaluation infrastructure. Accuracy was measured using the F1 score, a typical metric in the ML community.

The accuracy was compared against the object detection benchmark of 0.85, to indicate whether the pipeline was sufficiently precise for real-world application. Average precision was used to measure the relative performance of the model on each of the three classes.

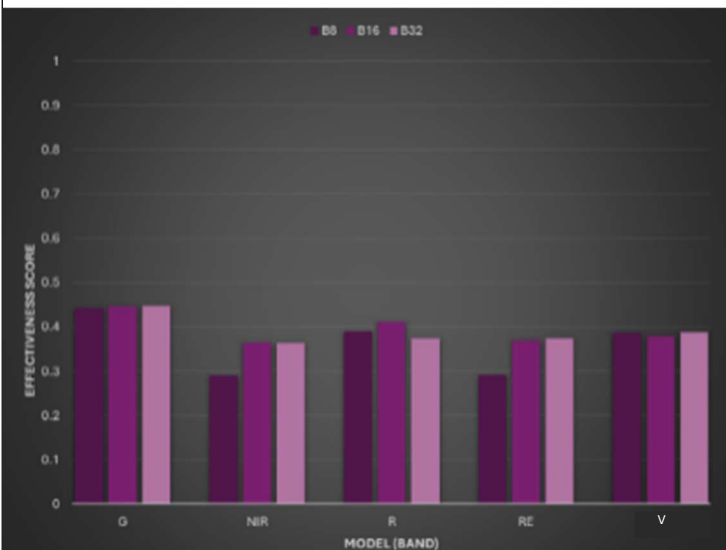
Results

PHASE I

Model Results

Figure 8 plots the effectiveness of each of the models across three different training iterations. Effectiveness serves as the cumulative indicator of model performance. Green had the highest average effectiveness of 0.445, followed by Red at 0.392, Visible at 0.385, Red-Edge at 0.345, and finally Near-InfraRed at 0.339. G, R, and V were consistently high, while NIR and NE suffered large decreases in their effectiveness.

Figure 8
Model Effectiveness



Note. G consistently shows greatest effectiveness; NIR consistently least optimal. Figure generated by S. Nair using Microsoft Excel, 2024.

Classification Results

G reported the highest true positive rate for bottles, with 204 bottles accurately predicted compared to NIR (141), R (184), RE (159), V (165). G reported the highest true positive rate for containers (49), compared to NIR (39), R (43), RE (35), V (46). No model predicted any Can object accurately. These results are summarized in Figure 9.

In terms of proportion, Can posed the greatest trouble for the detection models, as no object was correctly classified by any system. Container had a lower proportion of missed objects, followed by Bottle, which had the least proportion of missed objects across the models.

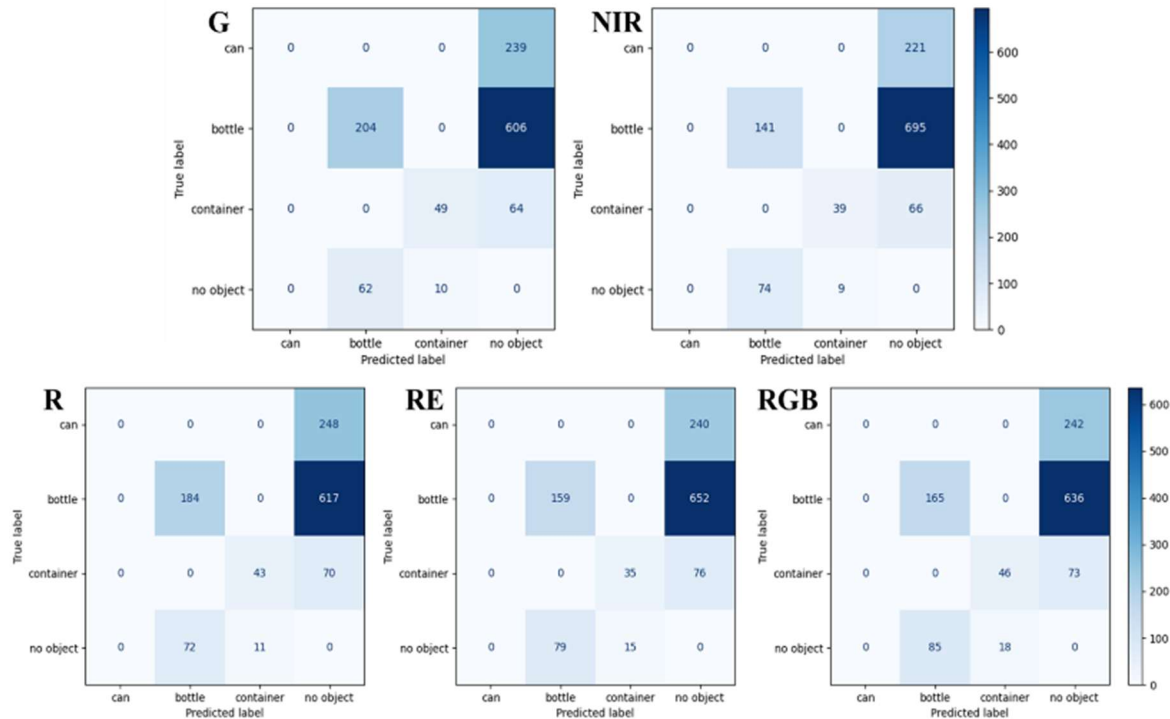
Dataset Statistical Analysis

Figure 10 visualizes the calculated pixel intensity ratios – between plastic pixels and background pixels – for G (2.51), NIR (1.53), R (1.88), RE (1.63), and V (1.45).

PHASE II

Machine Learning

Figure 9
Confusion Matrices Across Models



Note. a-b refers to the number of objects of class a that were predicted as b. Bottle-no object is consistently high; G has highest bottle-bottle; can-can is 0 across the board. Figure generated by S. Nair using Matplotlib (a Python publicly available graphing library), 2024.

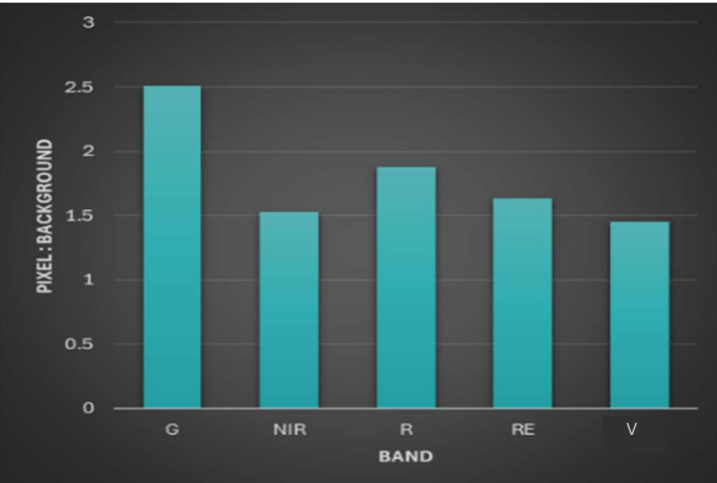
The total time consumption for training via RoboFlow was 1 hour and 28 minutes, a significant improvement from Phase I training time (17+ hours).

Model Evaluation & Statistical Analysis

The model finished with a precision of 88.8% and a recall of 83.0%. The F1 score was 0.8576. This value is greater than the benchmark of 0.85. Figure 11 compares the effectiveness of the Phase II model with those of the Phase I models.

Figure 10

Plastic to Background Pixel Intensity Ratios Across Bands



Note. Compared to G, NIR has a low ratio, though V is the lowest. G has the highest. Figure generated by S. Nair using Microsoft Excel, 2024.

The model performed most poorly on the *bottle* class (0.82), the most accurate on the *can* class (0.97), and its accuracy was 0.86 for the *container* class. Figure 12 displays these precisions alongside those of the top Phase I model (green).

Discussion

Phase I Model Results

Hypothesis

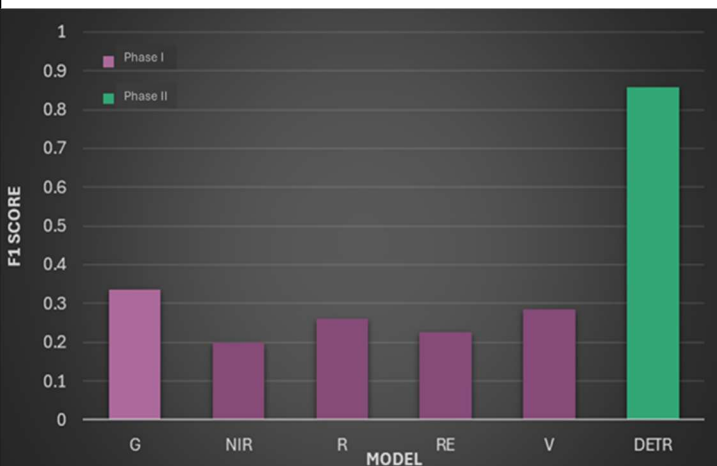
The hypothesis, that the Near Infrared model would demonstrate the highest effectiveness score, was *refuted* (* $p < .05$). The green band was significantly more optimal within the plastic identification task than the near infrared model, a result that is heretofore unreported in previous literature.

Effectiveness & Predictive Ability

The Phase 1 models had low effectiveness scores; none of the models reached the object detection benchmark

Figure 11

F1 Scores by Model in Phase II versus Phase I



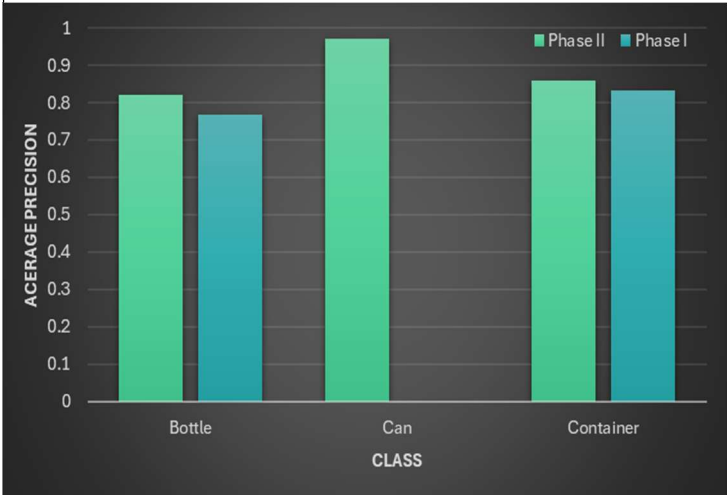
Note. The Phase II model (displayed in jade) outperformed all Phase I models (displayed in maroon), including the G model (displayed in pink). Figure generated by S. Nair using Microsoft Excel, 2025.

of 0.85. In fact, each of the models had a significantly high false negative rate. This implies that the models missed a large quantity of objects across the board, regardless of class.

According to the confusion matrix in Figure 9, all of the models had exceptionally poor performance with the can class. This could have been caused by many reasons, such as an error in the training architecture, corrupted output files from the jittering algorithm, or bad testing code. In response, the programming within Phase II was thoroughly tested to ensure that such bugs were not impacting the ML effectiveness.

Figure 12

Average Precision by Class in Phase II versus Phase I



Note. Phase II model had best performance in can, worst in bottle; showed greater precision than Phase I green model in all classes. Figure generated by S. Nair using Microsoft Excel, 2025.

Winans et al. (2023) noted that imbalances between class size – one or two classes having much more members than the others – resulted in poorer model predictions. The results suggest that similar issues have impacted the Phase I models. As depicted in Figure 13, the Container class was much more diverse than both the can and the bottle classes. The Container class was also much smaller in number of members than the other two classes, exacerbating the lopsidedness.

Regardless, it was evident that a robust detection pipeline would need a more accurate model.

Model Improvements in Phase II

After evaluating the Phase II framework, the null hypothesis was rejected, as the F1 score of the Phase II model – 0.86 – was greater than the benchmark F1 score of 0.85. This means that the percentage of correct predictions being made was suitable for the object detection quality threshold, verifying the pipeline as effective in real world use cases.

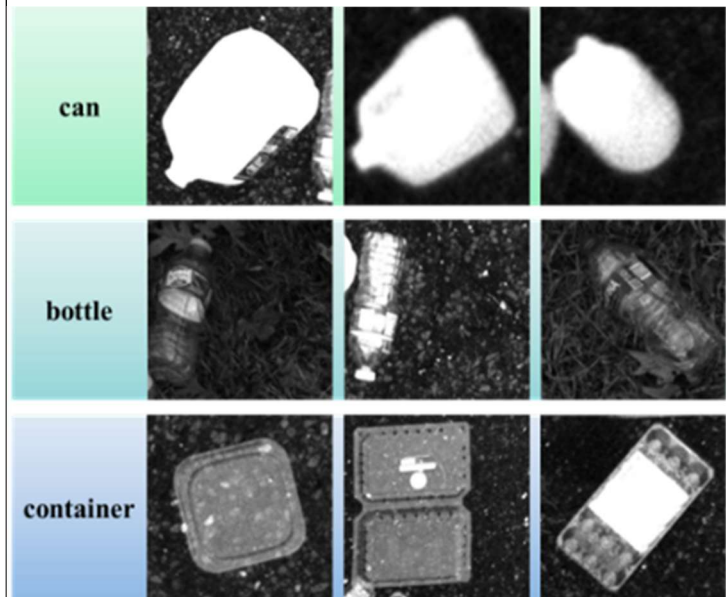
As the data used in Phase II was the same raw data from Phase I, the most impactful improvement was due to the choice of machine learning model. Phase I utilized the Single-Shot Multibox Detector paired with the MobileNet base architecture. While this pair had functioned well in previous literature (Winans et al., 2023), the size of the dataset in this study coupled with the challenges posed by the class imbalance of containers meant that the model was not able to reach a high accuracy. On the other

hand, the RF-DETR_medium that was used in Phase II was better suited for the challenges of this dataset. Its end-to-end approach meant that many of the complexities hindering MobileNet were mitigated; the process had less opportunity to overfit. In other words, the Phase II model was able to overcome dataset inconsistencies, implying that cleaner input data – more diverse, larger classes – would result in further advanced performance.

Another improvement came from the blurring augmentation step added to the preprocessing pipeline. This increased the number of images by threefold, allowing for the train-validation-test split to counteract the class size imbalance. Additionally, the blur mimicked the sacrifice of quality that occurs when one zooms in on an image; as many of the images require the model to make decisions on such “grainy” images, the blur augmentation assisted in training. This suggests that, to continue to improve the pipeline, other real-world impurities such as fogginess or clutter should be similarly mimicked in image augmentation preprocessing.

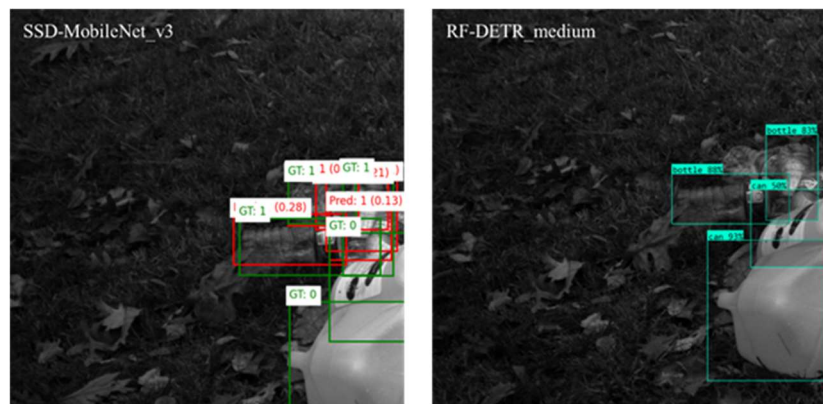
These improvements, coupled with my Phase I UniChip algorithm, allowed the Phase II model to outperform those of Phase I, which can be seen in Figure 14. Not only did the RF-DETR_medium locate all of the plastics in the image chip, it had greater confidence scores for each of its predictions than the SSD-MobileNet did (the former averaging around 86%; the latter averaging 30%). It also outperformed the model framework of Winans et al., (2023), who experimented with a similar task. As they did not

Figure 13
Variation Within Classes



Note. Can and Bottle classes have low variation; the Container class has high. Figure generated by S. Nair using Microsoft PowerPoint, 2024.

Figure 14
Phase I vs. Phase II Predictions



Note. Phase I (on the left) misses large can and has lower confidence. Phase II is more confident and accurate. Figure generated by S. Nair using Matplotlib, 2024.

have access to the DETR system, the imbalance in their classification scheme overpowered their SSD-MobileNet as well. Additionally, unlike Winans et al. (2023), this study optimized the mode of data collection by identifying the optimal qualities of the Green wavelength (560nm).

Green Band Dominance

Plastic Composition

Objects in the Bottle class consistently blended into the background, as depicted in Figure 15. These were commercial water bottles composed of PET, a thin-layered material. The objects were thus highly transparent to most forms of radiation. Despite the large quantity of Bottle objects, the accuracy of the models in both Phase I and Phase II towards identifying this class was not high. This issue could be tackled by experimenting with shorter wavelengths, such as ultraviolet, which may interact with PET at an atomic level more favorably than visible light, and reflect more strongly.

Objects in the Container class suffered a similar fate to those of the Bottle class. The class was highly diverse in shape, though all were composed of PET. Thus, they were easily overlooked due to their transparency and shape irregularity. However, by coupling the blur augmentation step to smarter processing systems, the Phase II model was able to overcome these obstacles, achieving a high accuracy with Container objects.

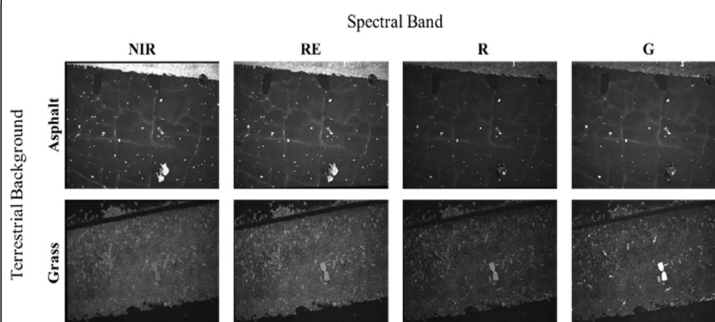
The Can class contained highly reradiative objects. As seen in Figure 15, every Can object was almost completely white in the Green band. This consistent reflectance was likely due to the HDPE composition. The polymer is inherently robust, chosen for applications that require toughness; thus, they are thicker and less transparent – more reflective – than the bottles and containers.

On average, however, all plastic classes demonstrated greater raw pixel intensity within the NIR images than within the G images. It suggests an internal trend between composition and spectral reradiation that should be investigated, to generalize to other plastic polymers. However, this also points to another factor driving the trend towards the short wavelengths: the reradiation of the background.

Background Composition

Figure 15

Leaf Litter on Grass and Asphalt Backgrounds in Multispectral Bands



Note. Leaf litter is more prominent on asphalt than on grass. Intensity within NIR and RE compared to plastic is much greater than within R and G. Figure generated by S. Nair using Microsoft PowerPoint, 2025.

The grass background had high reradiation across all five bands, but it was the brightest in NIR. This is consistent with plant behavior and the role of NIR. To optimize energy efficiency and reduce overheating, plants have evolved to absorb light in the blue and red wavelengths, with minimal absorption of green wavelengths. However, near-infrared is more adamantly scattered, as high NIR absorption would result in

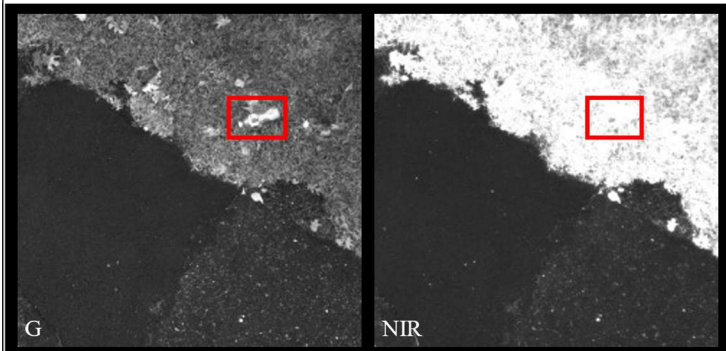
greater internal temperatures and harm to metabolism (Pettorelli et al., 2005). Because of this, the intensity of NIR radiation reflection by plant matter is much greater than that of even the Green band.

On the contrary, the asphalt background was consistently dim. Physically, asphalt has a rough texture, amplifying the absorption of radiation and reducing the amount measured by the camera. Chemically, asphalt is composed of long-chain hydrocarbons with delocalized electrons; this enhances absorption of infrared and visible light radiation. These two factors resulted in Asphalt's consistent spectral signature across all bands: it took on similar shades of dark grey and black. Overall, asphalt did not impact the detection abilities of the models, nor the efficacy of any particular spectral band.

As data collection for Phase I took place during late fall, miscellaneous dead organic matter – leaf litter, twigs – was present on both terrestrial backgrounds. While the twigs were largely invisible (they are not reflective in virtually any spectral band), the leaves posed a significant distraction to the NIR and RE models, as depicted in Figure 15. This is because of the internal structure of dry leaves: as they still have cell walls intact, and have a highly irregular overall form, they scatter more of the NIR radiation than grass. On the contrary, the chlorophyll that scatters green radiation entering grass is reduced in dead leaves, resulting in a comparatively small increase in reflectance in the green band.

These factors all contributed to the green band's greater plastic-to-background pixel intensity ratio (as seen in Figure 10), resulting in greater differentiation between the targeted objects and the background noise for the models (as seen in Figure 16), and thus a highly effective Green-based detection pipeline in Phase II. In other words, because the plastics in the green band images had greater contrast with the background than the plastics in the near infrared band images, the models were more effective at detecting and classifying them when using the green band. However, it was only until the implementation

Figure 16
Grass and Asphalt Backgrounds within G and NIR



Note. Both red bounding boxes contain the same plastic object; it is much more obscured in NIR than in G. Asphalt remains similar. Figure generated by S. Nair using Microsoft PowerPoint, 2025.

of DETR in Phase II that the pipeline could overcome the spectral obstacles posed by the Bottle transparency, and the small class size of Container.

Impact / Future

Development of a Plastic Detection Framework

The work conducted in Phase I refuted the hypothesis supported by previous research that the Near Infrared image band is the most optimal for plastic detection tasks.

The work conducted in Phase II demonstrated that a robust machine learning framework can be developed to tackle large-area terrestrial plastic detection by utilizing the green spectral band. By achieving an F1 score greater than 0.85, it suggested that the pipeline thus far is ready to be implemented into real-world applications involving drone mapping.

Creating an End-To-End Pipeline

In future research, the current system would be expanded to an end-to-end pipeline – meaning, one which would connect the drone imagery directly to a mapping visual tool. This would require steps to handle the DJI Mavic 3M image metadata, as well as the raw output of the machine learning framework, to create a geo-spatial mapping of an area of terrestrial surface in real time.

The transition from the existing framework to the proposed pipeline may necessitate additional tuning. For one, future research should consider the impact of continuous drone flight on the captured images. As the current framework was trained on images taken statically, the model may need a more involved preprocessing subsystem or motion-blur data augmentation steps to overcome complications arising from flying forward and capturing image data simultaneously.

Deploying the Pipeline

Previous studies of waste-management system improvements have documented cost reductions between 20 and 30%, especially following sensor-based monitoring (Barth et al., 2023). Based on recent estimates, the U.S. spends over \$11.5 billion annually on litter abatement (Wachter, 2024). Factoring in environmental hurdles such as terrain variation conservative estimations for cost reduction through implementing the pipeline in the Lower Hudson Watershed fall between 10-20%. Still, given the sheer cost of manual litter collection and scale of the pollution issue, this would nonetheless translate to substantial savings and a positive reallocation of resources.

Conclusion

In this study, I set out to bridge the gap between large-scale data collection methodologies and plastic presence data. By utilizing separate image masks from each of four spectral bands and one visible light image, coupled with a machine learning classification task, a novel result was supported: the green spectral band is more effective within a terrestrial plastic identification pipeline than the near-infrared band. This was likely due to the intense scattering of near-infrared radiation by the background as opposed to green radiation. The difference resulted in more pronounced ground-truth plastic objects in the green dataset than in the near-infrared dataset. In Phase II, I was able to construct a plastic detection framework by leveraging the green band optimization, and the resulting model passed the object detection proficiency benchmark. This suggests that the results of the study can feasibly be applied to real-world terrestrial plastic analysis tasks, by utilizing the plastic detection framework to gather location data over large areas. Through further research into the impacts of factors such as motion blur and interfering organic matter, a robust remote sensing plastic detection pipeline can be developed to construct plastic litter mappings at scale, enabling targeted and data-driven environmental management. By continuing to iterate and improve, this detection pipeline and the investigations that led to it can transform how municipalities and research institutions track, predict, and ultimately mitigate plastic pollution across New York's watersheds.

Supplemental Details

Software Specifications

Programming

All custom scripts and programs used were created in Visual Studio Code 1.98.1, using Python 3.12.5. Additionally, the following external Python modules were utilized: collections, csv, cv2, glob, itertools, matplotlib, multiprocessing, numpy, os, pandas, pickle, piexif, PIL, random, rawpy, scipy, seaborn, shap, shutil, skimage, sklearn, sys, time, torch, torchvision, tqdm, warnings, xml.

Data

Concerning all media and files necessary for this project, the total data size was 25.8 gigabytes. The data was handled on two Windows-11 OS devices; one using an Intel i7-1355u CPU, and the other using an RTX 4090 GPU. The following are all datatypes handled in this project: Python (.py), CSV (.csv), XML (.xml), HTML (.html), PTH (.pth), DNG (.dng), JPG (.jpg), TIF (.tif).

Formulae

Machine Learning

These are formulae involved with machine learning evaluation metrics, where E stands for Effectiveness, A stands for Accuracy, P stands for Precision, R stands for Recall, and F stands for F1 score.

$$(1) \quad E = w_a \times A + w_p \times P + w_r \times R + w_f \times F,$$

where $w_a = 0.1$, $w_p = 0.25$, $w_r = 0.25$, $w_f = 0.4$.

$$(2) \quad F = \frac{2 \times P \times R}{P + R}$$

Pipeline

These are formulae involved with high-level pipeline calculations, where v_{max} signifies maximum linear velocity of the drone in meters per second, H represents the vertical height of the drone off of the terrestrial surface in meters, θ represents the angle of the camera's field of view in radians, and S represents the inference time of each image in seconds.

$$(3) \quad v_{max} = \frac{2H \tan\left(\frac{\theta}{2}\right)}{S} \quad (4) \quad S = \frac{2H \tan\left(\frac{\theta}{2}\right)}{v_{max}}$$

References

- Barboza, L. G. A., Dick Vethaak, A., Lavorante, B. R. B. O., Lundebye, A.-K., & Guilhermino, L. (2018). Marine microplastic debris: An emerging issue for food security, food safety and human health. *Marine Pollution Bulletin*, 133(0025-326X), 336–348. <https://doi.org/10.1016/j.marpolbul.2018.05.047>
- Barth, L., Schweiger, L., Benedech, R., & Matthias Ehrat. (2023). From data to value in smart waste management: Optimizing solid waste collection with a digital twin-based decision support system. *Decision Analytics Journal*, 9, 100347–100347. <https://doi.org/10.1016/j.dajour.2023.100347>
- Cox, K. D., Covernton, G. A., Davies, H. L., Dower, J. F., Juanes, F., & Dudas, S. E. (2019). Human Consumption of Microplastics. *Environmental Science & Technology*, 53(12), 7068–7074. <https://doi.org/10.1021/acs.est.9b01517>
- Gandhi, N., Farfaras, N., Linda Wang, N.-H., & Chen, W.-T. (2021). Life Cycle Assessment of Recycling High-Density Polyethylene Plastic Waste. *Journal of Renewable Materials*, 9(8), 1463–1483. <https://doi.org/10.32604/jrm.2021.015529>

- Goyetche, R., Kortazar, L., & Amigo, J. M. (2023). Issues with the detection and classification of microplastics in marine sediments with chemical imaging and machine learning. *TrAC Trends in Analytical Chemistry*, 166, 117221. <https://doi.org/10.1016/j.trac.2023.117221>
- Lin, L., Di, L., Zhang, C., Guo, L., Zhao, H., Islam, D., Li, H., Liu, Z., & Middleton, G. (2024). Modeling urban redevelopment: A novel approach using time-series remote sensing data and machine learning. *Geography and Sustainability*. <https://doi.org/10.1016/j.geosus.2024.02.001>
- Pettorelli, N., Vik, J. O., Mysterud, A., Gaillard, J.-M., Tucker, C. J., & Stenseth, N. Chr. (2005). Using the satellite-derived NDVI to assess ecological responses to environmental change. *Trends in Ecology & Evolution*, 20(9), 503–510. <https://doi.org/10.1016/j.tree.2005.05.011>
- Raffaele Marfella, Prattichizzo, F., Celestino Sardu, Fulgenzi, G., Graciotti, L., Spadoni, T., Nunzia D’Onofrio, Scisciola, L., Rosalba La Grotta, Chiara Frigé, Pellegrini, V., Maurizio Municinò, Siniscalchi, M., Spinetti, F., Vigliotti, G., Vecchione, C., Albino Carrizzo, Giulio Accarino, Squillante, A., & Spaziano, G. (2024). Microplastics and Nanoplastics in Atheromas and Cardiovascular Events. *The New England Journal of Medicine*, 390(10), 900–910. <https://doi.org/10.1056/nejmoa2309822>
- Sivakumar, M., Roy, P., Harmsen, K., & Saha, S. (2003). *Satellite Remote Sensing and GIS Applications in Agricultural Meteorology World Meteorological Organization (WMO) India Meteorological Department (IMD) Centre for Space Science and Technology Education in Asia and the Pacific (CSSTEAP) Indian Institute of Remote Sensing (IIRS) National Remote Sensing Agency (NRSA) and Space Application Centre (SAC)*. https://www.unisdr.org/files/1682_9970.pdf
- Tedesco, M., & Radzikowski, J. (2023). Assessment of a Machine Learning Algorithm Using Web Images for Flood Detection and Water Level Estimates. *GeoHazards*, 4(4), 437–452. <https://doi.org/10.3390/geohazards4040025>
- Tsuchiya, M., Tomo Kitahashi, Nakajima, R., Kazuta Oguri, Kawamura, K., Nakamura, A., Nakano, K., Maeda, Y., Murayama, M., Chiba, S., & Katsunori Fujikura. (2024). Distribution of microplastics in bathyal- to hadal-depth sediments and transport process along the deep-sea canyon and the Kuroshio Extension in the Northwest Pacific. *Marine Pollution Bulletin*, 199, 115466–115466. <https://doi.org/10.1016/j.marpolbul.2023.115466>
- Vethaak, A. D., & Leslie, H. A. (2016). Plastic Debris Is a Human Health Issue. *Environmental Science & Technology*, 50(13), 6825–6826. <https://doi.org/10.1021/acs.est.6b02569>

- Wachter, H. (2024, March 25). *The High Cost of Cleaning Up Litter*. Experience Life.
https://experiencelife.lifetime.life/article/the-high-cost-of-litter/?utm_source=chatgpt.com
- Wen, Z., Wang, Q., Ma, Y., Jacinthe, P. A., Liu, G., Li, S., Shang, Y., Tao, H., Fang, C., Lyu, L., Zhang, B., & Song, K. (2023). Remote estimates of suspended particulate matter in global lakes using machine learning models. *International Soil and Water Conservation Research*.
<https://doi.org/10.1016/j.iswcr.2023.07.002>
- Winans, W. R., Chen, Q., Qiang, Y., & Franklin, E. C. (2023). Large-area automatic detection of shoreline stranded marine debris using deep learning. *International Journal of Applied Earth Observation and Geoinformation*, 124, 103515. <https://doi.org/10.1016/j.jag.2023.103515>
- Wright, S. L., Thompson, R. C., & Galloway, T. S. (2013). The physical impacts of microplastics on marine organisms: A review. *Environmental Pollution*, 178(178), 483–492.
<https://doi.org/10.1016/j.envpol.2013.02.031>
- Wu, W.-M., Yang, J., & Criddle, C. S. (2016). Microplastics pollution and reduction strategies. *Frontiers of Environmental Science & Engineering*, 11(1). <https://doi.org/10.1007/s11783-017-0897-7>
- Zhang, J., Zou, T., & Lai, Y. (2021). Novel method for industrial sewage outfall detection: Water pollution monitoring based on web crawler and remote sensing interpretation techniques. *Journal of Cleaner Production*, 312, 127640–127640. <https://doi.org/10.1016/j.jclepro.2021.127640>

